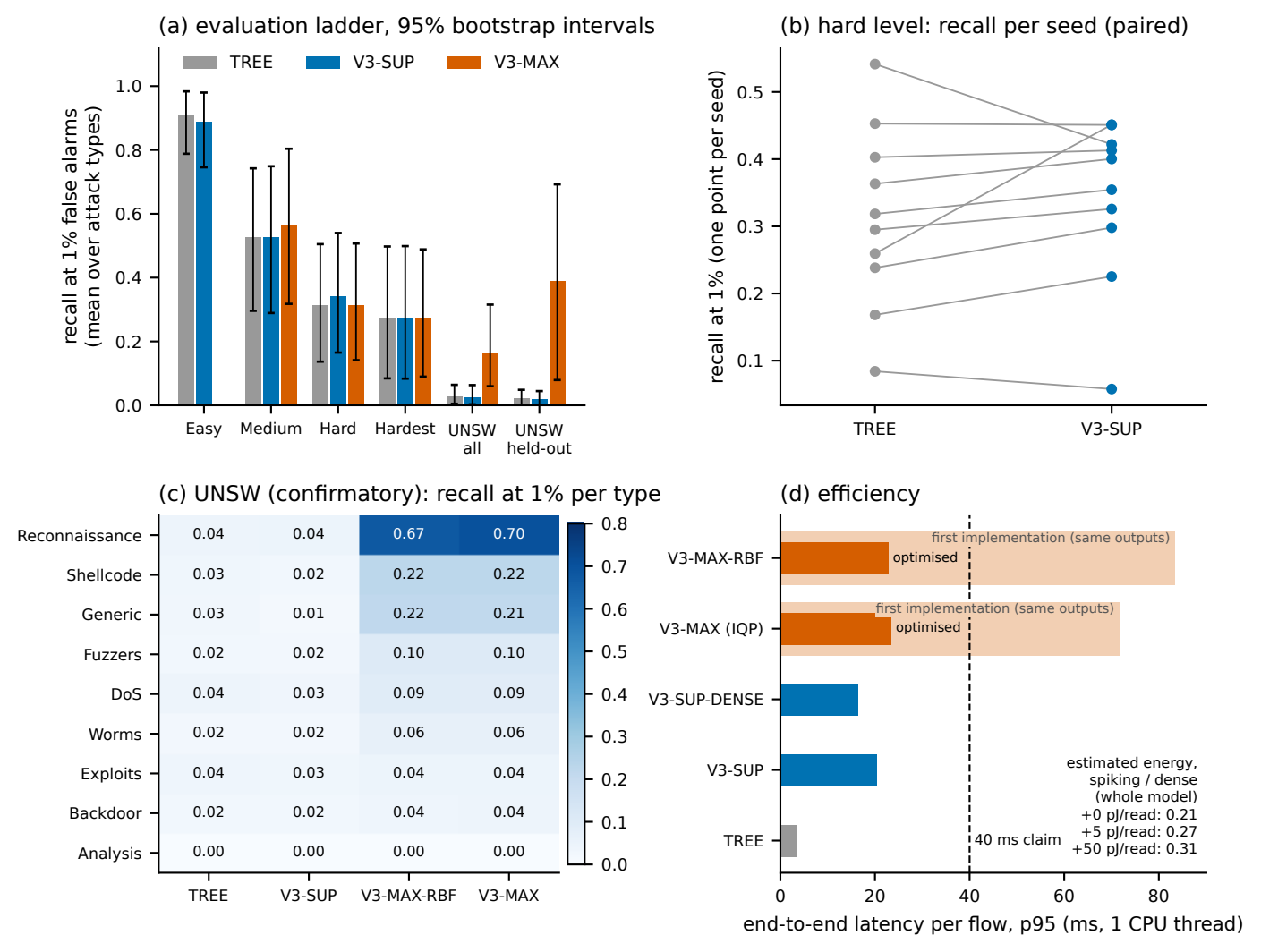
(a) evaluation ladder, 95% bootstrap intervals
TREE
V3-SUP
V3-MAX
recall at 1% false alarms
(mean over attack types)
1.0
0.8
0.6
0.4
0.2
0.0
Easy
Medium
Hard
Hardest
UNSW all
UNSW held-out
(b) hard level: recall per seed (paired)
recall at 1% (one point per seed)
0.5
0.4
0.3
0.2
0.1
TREE
V3-SUP
(c) UNSW (confirmatory): recall at 1% per type
Reconnaissance 0.04 0.04 0.67 0.70
Shellcode 0.03 0.02 0.22 0.22
Generic 0.03 0.01 0.22 0.21
Fuzzers 0.02 0.02 0.10 0.10
DoS 0.04 0.03 0.09 0.09
Worms 0.02 0.02 0.06 0.06
Exploits 0.04 0.03 0.04 0.04
Backdoor 0.02 0.02 0.04 0.04
Analysis 0.00 0.00 0.00 0.00
TREE
V3-SUP
V3-MAX-RBF
V3-MAX
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
(d) efficiency
first implementation (same outputs)
optimised
V3-MAX-RBF
first implementation (same outputs)
optimised
V3-MAX (IQP)
V3-SUP-DENSE
V3-SUP
TREE
40 ms claim
estimated energy,
spiking / dense
(whole model)
+0 pJ/read: 0.21
+5 pJ/read: 0.27
+50 pJ/read: 0.31
0
20
40
60
80
end-to-end latency per flow, p95 (ms, 1 CPU thread)